

Toolbox

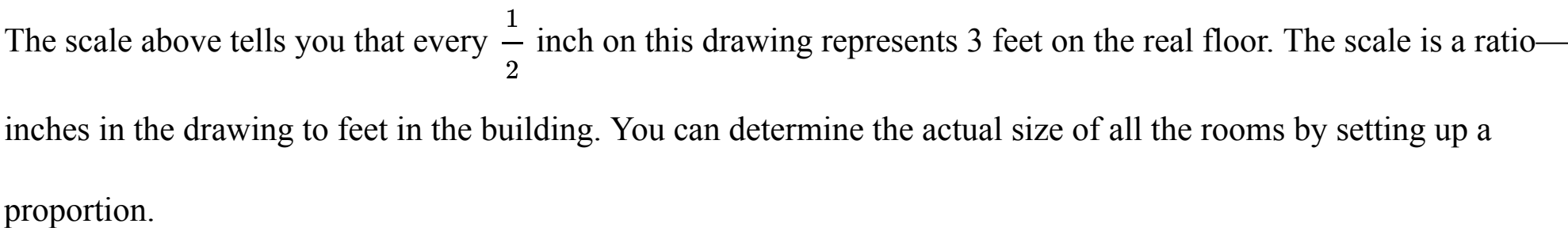
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Lesson 6: Ratios and Proportions

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The window in the drawing measures $\frac{3}{4}$ inch. How many feet across will the bay window be?

Step 1

Set up a proportion.

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$$\frac{\frac{1}{2} \text{ inch}}{3 \text{ feet}} = \frac{\frac{3}{4} \text{ inch}}{n \text{ feet}} \rightarrow \left(\frac{1}{2}\right) \times n = \left(\frac{3}{4}\right) \times 3 \rightarrow \frac{n}{2} = \frac{9}{4}$$

Step 2

Multiply to find the cross products and find the unknown.

$$\frac{1}{2} \times \frac{3}{4} \times 3 = \frac{9}{8}$$

$$\frac{1}{2} \times \frac{1}{n} = 2 \frac{1}{4}$$

$$n = \frac{1}{4^2} \text{ feet}$$

Step 3

Check your proportion by using cross products.

$$\frac{\frac{1}{2}}{3} = \frac{\frac{3}{4}}{4\frac{1}{2}}$$

$$\frac{1}{2} \times 4\frac{1}{2} = \frac{3}{4} \times 3$$

$$2^{\frac{1}{4}} = 2^{\frac{1}{4}}$$

The bay window will be $4\frac{1}{2}$ feet across.